HACKATHON PRESENTATION 2026

GapLens

AI-Assisted Epistemic Gap & Void Detection Platform

The Paradigm Shift:

"Every current AI summarizes what has been written. GapLens exposes the critical omissions, conflicting claims, and temporal decay that will get you blindsided."

TRACK

Epistemic AI & Information Reliability

CORE VALUE

Auditing Absence Over Retrieval

GITHUB REPOSITORY

github.com/hemantpatil225577/gaplens

THE CORE PROBLEM

The Digital Dark Spot & Retrieval Blindness

THE RETRIEVAL BLINDNESS TRAP

- **False Sense of Completeness:**
  Users assume a well-crafted AI answer covers everything that matters. In reality, LLMs only synthesize ingested text without checking for missing conditions.
- **Absence is Completely Invisible:**
  Standard RAG only retrieves keyword matches. If an applicant doesn't know to ask about "retroactive clawbacks", neither Google nor ChatGPT will alert them.
- **Silent Catastrophes in High Stakes:**
  Students lose $80k scholarships over unlisted GPA maintenance rules; patients enter trials missing exclusion criteria; startups sign MSAs with unmentioned 4-hour SLA downtime exclusions.

Epistemic Omission (Dark Spots)

HIGH RISK

Critical prerequisites, hidden GPA thresholds, and clawback liabilities omitted across all consulted web portals.

Unstated Contradictions

AMBIGUITY

Disparate portals quote mutually exclusive thresholds (e.g. $45k vs $60k income ceiling or 1 vs 2 recommendation letters).

Temporal Drift & Decay

STALE DATA

Outdated 2023 handbooks and archived forum advice surface as top search hits and are erroneously applied to 2026 cycles.

THE SOLUTION

GapLens: Modeling the Negative Space of Knowledge

01

Auditing Absence

Epistemic Negative Space

- Analyzes what is NOT said across the entire corpus of provided sources.
- Audits against domain ontologies (Scholarships, Grants, Visas, Medical, Legal).
- Detects hidden disqualifiers, clawbacks, and unmentioned prerequisites.

02

Forensic Divergence

Verbatim Contradictions

- Performs cross-source sentence alignment to spot conflicting numbers and rules.
- Extracts exact side-by-side verbatim quotes with source attribution.
- Provides transparent guidance on which authority takes legal precedence.

03

Actionable Verification

Pre-Commitment Punchlist

- Transforms discovered voids into a prioritized verification checklist.
- Routes each uncertainty to the exact office to call (bursar, consul, registrar).
- Includes a "What-If" stress testing simulator to forecast decision failure modes.

SYSTEM ARCHITECTURE

Dual-Pass Epistemic Pipeline & Provenance Engine

STEP 1

Ingest & Clean

Raw text, PDF, URL proxy with Cheerio DOM cleaning and date metadata heuristics

STEP 2

Epistemic Audit

Gemini 3.8 Flash audits 6 dimensions: eligibility, clawbacks, renewal, deadlines, liabilities

STEP 3

Divergence Map

Cross-document alignment identifies semantic conflicts & temporal cycle mismatches

STEP 4

Action Dossier

Calculates completeness score, builds verification checklist & stress-test simulator

CORE TECHNICAL INNOVATIONS

- **Epistemic Dimension Matrix:**
  Evaluates omissions across 6 structural axes: Eligibility, Financials, Renewal Rules, Deadlines, Jurisdiction, and Liabilities.
- **Strict JSON Schema Enforcement:**
  Calls Gemini with responseMimeType "application/json" and rigid response schemas for zero hallucinated formatting.
- **Verbatim Citation Provenance:**
  Zero synthetic quotes; all contradictions extract exact text snippets directly from the ingested source documents.

SECURITY & FULL-STACK RESILIENCE

- **Server-Side Proxy Architecture:**
  Node.js Express backend keeps Gemini API credentials isolated. Client applications never expose private API keys.
- **Precomputed High-Stakes Benchmarks:**
  Instant demo execution across four distinct real-world domains with zero latency or cold-start delay.
- **Deterministic Heuristic Fallback Engine:**
  Built-in offline discrepancy and omission rules guarantee 100% demo uptime even under network or rate-limit constraints.

KEY FEATURES

Built for High-Stakes Decision Verification

VOID DETECTION

Dark Spot Matrix & Topology

Visualizes missing facts by category and severity. Displays why each omission is hazardous and what specific questions must be asked.

CONFLICT RADAR

Contradiction Forensic Radar

Side-by-side claim comparison with exact quotes, divergence analysis, and guidance on which policy or year takes legal precedence.

TEMPORAL DRIFT

Temporal Decay & Cycle Audit

Detects outdated documentation (e.g. 2023 regulations applied to 2026 applications) and flags stale terms vulnerable to policy drift.

STRESS TESTER

Interactive Decision Simulator

Lets users stress-test hypothetical scenarios: "What if my GPA drops to 3.4?" or "What if I take a leave of absence?" with real-time risk projections.

PROVENANCE

Verbatim Source Inspector

Full-text in-app document viewer with keyword search and highlight, proving zero hallucination and showing exact textual context.

EXPORT SUITE

Audio Briefing & Multi-Export

Generates browser text-to-speech voice debriefings and exports comprehensive audit dossiers to Markdown, JSON, and PowerPoint presentations.

COMPETITIVE EDGE

GapLens vs. Standard Search & Conventional LLM Summarizers

Capability / Metric	Google Search / Perplexity	ChatGPT / Standard RAG	GapLens Epistemic Auditor
Detection of Unstated Gaps	None (Only retrieves keywords)	Rare (Prone to hallucinating answers)	**Native (Epistemic Void Modeling)**
Cross-Source Contradictions	Left to human to compare	Smooths over discrepancies	**Exact verbatim quote alignment**
Temporal Staleness Audit	Ranked by SEO, not validity	No cycle or expiration check	**Flags cycle drift & legacy rules**
Decision Actionability	Unstructured web links	Generic paragraphs of advice	**Prioritized verification checklist**
Evidence Transparency	Web snippet only	Opaque neural generation	**Verbatim source inspection**

DEMO BENCHMARKS

Tested Across 4 Real-World High-Stakes Domains

HIGHER EDUCATION FUNDING

1. STEM Fellowship Trap

Dark Spot Omission:

Hidden 3.75 GPA requirement & $80,000 retroactive clawback upon changing majors.

Factual Divergence:

Flyer claims $80k income cap; Official Bylaws state $65k hard ceiling.

BIOMEDICAL & HEALTHCARE

2. Oncology Clinical Trial

Dark Spot Omission:

Unpublished autoimmune exclusion clause; lack of post-trial drug access guarantee.

Factual Divergence:

Hospital portal says travel is 100% subsidized; consent form specifies $500 cap.

B2B SOFTWARE & LEGAL

3. Cloud Enterprise MSA

Dark Spot Omission:

Zero compensation for outages under 4 hours; unmentioned 8% auto-renewal escalator.

Factual Divergence:

Marketing brochure claims 99.99% uptime; legal terms exclude scheduled maintenance.

IMMIGRATION & RELOCATION

4. Tech Work Visa Pathway

Dark Spot Omission:

Absence of spouse work authorization (H-4) rules & mandatory 6-month exit grace.

Factual Divergence:

Government portal lists $1,500 fee; embassy bulletin lists $2,800 processing fee.

ROADMAP & IMPACT

From Hackathon Prototype to Epistemic Infrastructure

PHASE 1 (TODAY)

Full-Stack Web Auditor

- Live Express + React + Gemini pipeline
- Multi-source ingestion & text scraper
- Interactive topology & blind spot probe
- Open-source repository on GitHub

PHASE 2 (Q2 2026)

Browser Real-Time Sentinel

- Chrome/Edge extension reading Terms of Service
- Real-time ambient dark spot alerts on sign-up pages
- Automatic cross-checking against official gazettes
- Crowdsourced community void reporting

PHASE 3 (Q4 2026)

Enterprise Epistemic API

- SDK integration for legal tech & compliance
- Automated contract red-flagging before signature
- Academic peer review omission scanner
- Enterprise data room completeness audit

TRY GAPLENS LIVE & EXPLORE THE OPEN SOURCE CODEBASE

GitHub: https://github.com/hemantpatil225577/gaplens

Licensed under Apache-2.0 • Built with Google Gemini 3.8 Flash & React 19 • Ready for evaluation